Coordinate-Governed Mapping of Source-Local Scientific Projections

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Abstract

Scientific knowledge integration can become overconfident when a mapping hides the coordinate that determines whether two source claims agree. We study a mapping rule for already-structured scientific projections that keeps referent, construct, measurement, context, polarity, modality and attribution conditions separate and permits an explicit non-merge outcome. In a 32-case, case-identifier-disjoint public-reference holdout frozen before confirmatory outputs were examined, coordinate-governed mapping made no false merges, compared with a false-merge rate of 0.1875 for flat predicate canonicalization. The paired difference was -0.1875 with a 95% bootstrap interval of $[-0.34375, -0.0625]$. The false-split difference relative to an exact-coordinate conservative control was 0.000 with interval $[0.000, 0.000]$, satisfying both conditions of the predeclared rule. A structurally separate scorer reproduced the case-level result from the frozen records. The evidence concerns structured mapping decisions only; raw-text extraction, expert-atlas generality and downstream scientific utility were not tested, and the observed errors were all polarity contrasts. Outcomes without sufficient evidence remain undetermined rather than being counted as negative results.

Keywords: scientific knowledge integration; semantic interoperability; provenance; ontology mapping; measurement harmonization.

1 Introduction

Scientific claims are distributed across disciplines with different terminologies, measurements, contexts and representational conventions. A global synthesis can therefore fail even when every local statement is well formed: identical terms can denote different objects, different labels can denote compatible objects, and apparent contradictions can arise from modality, attribution or context. This paper studies the downstream decision of whether two already-structured scientific representations may be treated as one statement. It does not study retrieval or raw-text extraction.

The proposed rule represents each source as a provenance-bound scientific projection. A candidate mapping records its referent, construct, measurement, context, polarity, modality and attribution conditions separately rather than reducing them to a single similarity score. Compatible projections may be integrated when the conditions required by the claim are preserved. A load-bearing incompatibility produces a typed non-merge outcome, while insufficient evidence leaves the relation undetermined.

We evaluate this rule on two 32-case public-reference mapping sets. The first set motivated a prospectively frozen replication. The second set shared no case identifiers with the first and was

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bound, together with its source revisions, evaluator identities, bootstrap seed, margins and terminal rule, before confirmatory outputs were examined. On the confirmatory holdout, coordinate-governed mapping made no false merges, whereas flat predicate canonicalization false-merged six of 32 cases. The paired false-merge difference was -0.1875 with a 95% bootstrap interval of $[-0.34375, -0.0625]$. The false-split difference against an exact-coordinate conservative control was zero, satisfying the non-inferiority guard specified in advance.

The contribution is deliberately bounded. The experiment tests mapping decisions over already-structured projections, and all six discriminating confirmatory cases were polarity contrasts. It does not establish raw-text integration performance, expert construct validity across a broader case universe, downstream answer-quality gains or the necessity or empirical value of every coordinate. Those questions require separate prospective evidence.

Section 2 positions the decision rule relative to scientific knowledge representation, schema matching, provenance and semantic interoperability. Sections 3–5 define the rule and the frozen evaluation. Section 6 reports the confirmatory result and secondary ablations. Sections 7 and 8 state the boundary and conclusion.

2 Related work

The evaluated decision lies downstream of extraction and knowledge-graph construction. MUSE recovers source-grounded problem–solution–rationale structures from scientific text [1], SciER annotates scientific entities and relations [16], and SciSchema.org supplies expert-reviewed process schemas with measurements, conditions and provenance fields [2]. SCOPE and SCION evaluate evidence-linked schema induction and optional conservative fusion [6], while Executable Schema Contracts uses discovered structure to guide provenance-aware integration and retrieval [8]. These systems establish strong upstream representations. Our experiment instead holds the structured projection fixed and evaluates the subsequent merge decision.

Ontology matching studies correspondences including equivalence, subsumption and disjointness [3]; recent systems such as LLMATCH add schema preparation, candidate selection and fine-grained alignment for complex schemas [15]. Reviews of knowledge-graph construction and scholarly knowledge graphs cover schema matching, entity resolution, fusion, provenance and refinement [5, 11]. The rule evaluated here does not replace those components. It asks whether a proposed correspondence preserves the scientific coordinates required by a particular claim, and records why a merge is permitted, refused or left undetermined.

Scientific-variable and interoperability research supplies especially close conceptual foundations. I-ADOPT decomposes observable properties into machine-readable components [10], and its recent benchmark shows the continuing difficulty of constructing such descriptions from scientific language [9]. FAIR 2.0 distinguishes several levels of semantic interoperability and emphasizes explicit mappings rather than a single privileged representation [13]. DEC interprets provenance-bearing statements as epistemic stance and preserves source-relative disagreement [12]. The present work combines these donor ideas only at the decision layer: coordinate conditions and provenance govern whether two structured projections can be integrated.

Measurement equivalence provides a further boundary. Nominally similar outcome measures need not be comparable across settings, and equivalence may require explicit testing [7]. Stance detection likewise shows that support and opposition are target-relative rather than context-free labels [4]. Treating measurement, polarity, modality, attribution and context as explicit mapping conditions makes those distinctions inspectable instead of hiding them in a scalar match score.

The residual contribution is therefore not a new extractor, ontology matcher, provenance model or universal integration architecture. It is a bounded, prospectively confirmed comparison of decision rules applied after scientific projections have already been structured. The nearest systems provide richer upstream construction and alignment capabilities; the present evidence adds a small claim-relative non-merge test and reports where that test has, and has not, been evaluated.

3 Coordinate-governed mapping

The implementation evaluated in this study is the ORION knowledge-integration layer. It is named here only to identify the artifact under test; the scientific object is the mapping rule defined below.

3.1 Three-valued decisions

A mapping can be determined admissible, determined inadmissible, or left *undetermined*. The third value is used when a required coordinate or source authority is absent. It is not counted as a negative decision and is retained separately in every aggregate.

3.2 Source-local projections

Let a source record s bind a document or structured source item to an exact identity, revision, locator and content commitment. Its scientific meaning is represented by

$$P_s = \langle R_s, C_s, M_s, X_s, \rho_s, \mu_s, a_s, \delta_s, \Pi_s \rangle, \tag{1}$$

where R_s denotes the referent, C_s the construct, M_s the measurement or operationalization, X_s the context, ρ_s polarity, μ_s modality, a_s attribution, δ_s discourse relation, and Π_s assumptions and undetermined information. Unsupported coordinates remain absent; the study does not infer missing values from raw papers.

3.3 Coordinate-governed mapping

For two projections, the rule evaluates a vector of typed coordinate relations rather than a scalar similarity score. A candidate mapping records the proposed relation, the coordinates it claims to preserve, the conditions required for use, and any information it does not preserve. A mapping is admissible only if the relations required by the claim are compatible and the preservation conditions hold. A required incompatibility produces a typed non-merge record naming the failed coordinate; insufficient evidence leaves the mapping undetermined.

The confirmatory experiment tests this decision semantics. Its observed discrimination occurs in six polarity contrasts, where the flat comparator merges predicates without the additional condition. The experiment does not identify the independent necessity or value of the other coordinates.

3.4 Recoverable provenance

Each case retains the external source identity and the frozen derivation of its expected mapping decision. Each output retains the source records and coordinate conditions that support the decision. This design permits a structurally separate implementation to reconstruct the comparison from the frozen records without treating a digest or repository receipt as scientific evidence by itself.

Confirmatory case family	Cases
Different name / same referent	13
Polarity, modality, attribution, or context	13
Valid / invalid representation mapping	6
Total	**32**

Table 1: Composition of the case-identifier-disjoint confirmatory public-reference holdout.

4 Public-reference mapping datasets

The study uses two content-addressed public-reference mapping datasets. They contain already-structured scientific projections assembled from pinned MUSE, SciFact [14] and SciSchema resources and from deterministic standards. They are not newly commissioned expert annotations and do not evaluate extraction from raw text.

The initial set contains 32 cases and is used only as development evidence. A second 32-case holdout was selected by an outcome-blind round-robin window over the sorted source pools, beginning after the initial window and excluding every initial case identifier. The case-identifier overlap is zero. Two source-locator–content-hash records recur across the two sets, so the holdout is not source-disjoint. Before confirmatory outputs were examined, an execution manifest froze the holdout digest, source revisions, evaluator identities, bootstrap seed, decision margins and terminal rule.

The different-name/same-referent cases come from the MUSE cross-domain stratum. The polarity, modality, attribution and context cases come from a scientific claim-verification stratum. The representation-mapping cases span materials, physics and psychology and use deterministic standards where appropriate. The confirmatory set is most discriminating in the second family: flat predicate canonicalization false-merges six polarity-contrast cases among its 13 cases, whereas every compared rule is correct in the other two families.

A broader eight-family expert-atlas protocol is retained as prospective work. Its templates are not expert gold: no independent dual annotation, specialist adjudication or prospectively bound raw-text model run exists for that object. It is excluded from the present claim rather than being redescribed as the dataset used.

5 Evaluation

The confirmatory protocol was frozen after the initial 32-case result but before any output on the case-identifier-disjoint holdout was examined. The initial result could motivate replication but could not change the holdout, evaluator, margins or pass rule.

5.1 Primary decision rule

The hypothesis had two non-compensatory conditions. First, the coordinate-governed-minus-flat false-merge estimate had to be at most -0.05, with the upper bound of its paired 95% bootstrap interval below zero. Second, the coordinate-governed-minus-exact-conservative false-split estimate had to be at most $+0.03$, with the interval upper bound no greater than $+0.03$. The holdout passed only if both conditions held.

5.2 Comparators

All three rules received the same structured cases. The coordinate-governed rule evaluated the required scientific coordinates and could retain a typed non-merge decision. Flat predicate canonicalization treated compatible predicates as sufficient merge evidence and was the predeclared false-merge comparator. The exact-coordinate conservative control required exact coordinate compatibility and could abstain; it was the predeclared false-split control. These are mapping-layer comparisons, not reproductions of complete external knowledge-graph or retrieval systems.

5.3 Metrics and inference

The scientific unit is one frozen mapping case. Primary outcomes are the case-level false-merge and false-split rates. Individual rates use Wilson 95% intervals. Paired differences use 10,000 percentile-bootstrap resamples with seed 20260817. The rules are deterministic, so one decision per case is used rather than artificial stochastic repetitions. The initial 32 cases are not pooled with the confirmatory holdout.

Secondary ablations force compatibility without the non-merge state or remove coordinate groups. Their paired intervals are descriptive. They cannot create new confirmatory claims, and a zero effect on this holdout is not interpreted as evidence that a coordinate is unnecessary.

5.4 Fail-closed replay

Any mismatch in the frozen gold digest, source revisions, evaluator identities or case overlap leaves the replication undetermined rather than silently changing the study. The expected decisions are withheld from the mapping rule and are opened by the evaluator only after candidate outputs have been sealed.

6 Results

6.1 Initial mapping result

On the initial 32 cases, coordinate-governed mapping had accuracy 1.000, with false-merge and false-split rates of 0.000. Flat predicate canonicalization had accuracy 0.875 and a false-merge rate of 0.125. The paired false-merge difference was -0.125 with a 95% interval of $[-0.250, -0.03125]$. The exact-coordinate conservative control made no false merges or false splits but abstained on 0.125 of cases. This result motivated the replication and is not pooled into its verdict.

6.2 Prospectively frozen replication

On the confirmatory holdout, coordinate-governed mapping made no false merges or false splits. Flat predicate canonicalization false-merged six of 32 cases and had accuracy 0.8125. The paired false-merge difference was -0.1875 with a 95% bootstrap interval of $[-0.34375, -0.0625]$. The false-split difference against the exact-coordinate conservative control was 0.000, with interval $[0.000, 0.000]$. Both predeclared conditions were therefore satisfied.

The effect was localized. Flat predicate canonicalization false-merged six polarity-contrast cases in the 13-case polarity/modality/attribution/context family and none of the other 19 cases. All three rules were correct in the different-name/same-referent and representation-mapping families. The result therefore supports the tested polarity distinction; it does not establish the necessity or empirical value of every coordinate.

Rule	Accuracy	False merge	False split	Abstention
Coordinate-governed mapping	1.0000	0.0000	0.0000	0.0000
Flat predicate canonicalization	0.8125	0.1875	0.0000	0.0000
Exact-coordinate conservative	0.8125	0.0000	0.0000	0.1875

Table 2: Confirmatory outcomes on 32 frozen mapping cases. Rates use one decision per case. The paired false-merge difference for coordinate-governed minus flat mapping was -0.1875 (95% bootstrap interval [-0.3438, -0.0625]). The paired false-split difference relative to the exact-coordinate control was 0.0000 ([0.0000, 0.0000]).

6.3 Secondary ablations and null results

Forcing compatibility without the explicit non-merge state increased the false-merge rate by +0.1875 with a 95% paired interval of $[+0.0625, +0.34375]$. Removing the modality, polarity, attribution and discourse group produced the same change. Removing referent, construct, measurement or temporal context produced a measured difference of zero on this holdout. These null effects are retained as coverage limits, not as evidence that the coordinates are dispensable.

6.4 Separate-implementation check

A standard-library-only scorer that imports none of the original analysis code reproduced the 32-case decisions, primary differences and verdict from the frozen records. It also confirmed zero case-identifier overlap and two recurring source-locator–content-hash records. This is a structurally separate implementation check within the same repository snapshot and custody, not an external or cross-host replication.

7 Limitations and interpretation

The evidence supports a narrow mapping claim. The confirmatory sample contains 32 already-structured cases from three case families, and all observed discrimination comes from six polarity contrasts. Although case identifiers do not overlap the initial set, two source-locator–content-hash records recur. The study does not estimate prevalence or error rates in an unrestricted population of scientific claims.

The flat comparator is intentionally weak and isolates the failure caused by predicate-only merging. Strong schema-induction, ontology-matching and provenance-aware systems were studied as nearest work but were not executed as same-universe comparators. The experiment therefore establishes an advantage over the registered decision rule, not superiority over current integration systems.

The source records reuse public human or expert resources, but the broader eight-family expert atlas has not been executed. Raw-text extraction, specialist construct validation, generated-portrait recoverability, naturalistic cross-domain transport and downstream scientific-answer quality remain undetermined. A structurally separate replay checks the frozen computation, but it shares repository custody and is not independent external replication or review.

Finally, zero ablation effects for referent, construct, measurement and temporal context reflect missing comparison opportunity in this holdout. They do not show that those coordinates are unnecessary. A future test of coordinate necessity requires cases in which each coordinate contrasts within otherwise comparable pairs, together with stronger same-universe comparators and independent evaluation.

8 Conclusion

For the frozen public-reference mapping task, keeping polarity explicit in the registered coordinate rule prevented six false merges made by flat predicate canonicalization while satisfying the predeclared false-split guard against a conservative control. A separate implementation reproduced the bounded result from the frozen case records.

The conclusion stops at that mapping layer. It does not establish raw-text integration performance, general superiority over semantic-integration systems, expert-atlas validity, downstream utility or the necessity of every coordinate. Within the tested scope, the result supports a simple discipline: integrate source-local projections only when the scientific conditions required by the claim are preserved, and keep incompatibility or insufficient evidence visible rather than forcing a canonical merge.

9 Data and code availability

The public-reference case records, source registry, frozen confirmatory protocol, deterministic analysis, generated tables and separate-implementation verification record are included in the versioned research package supporting this manuscript. The archived package preserves version and integrity information for the initial and confirmatory case records. Deterministic instructions regenerate the analysis and publication tables from these records.

 Restricted third-party text is not redistributed; the package retains source identity, revision, locator and content commitment. The exact review package must be deposited at a long-term stable, immutable URL before submission. No such URL is asserted in this draft.

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